

Siri Chandana Bollimpalli — AI/ML Engineer

(404) 981-1138 | siribollimpalli31@gmail.com

PROFESSIONAL SUMMARY:

- Around 4 years of experience in Artificial Intelligence and Machine Learning developing data-driven solutions using machine learning, deep learning, and statistical modeling techniques.
- Experienced in Python-based machine learning development, predictive analytics, model experimentation, and building AI solutions for healthcare, financial services, and e-commerce domains.
- Strong understanding of machine learning frameworks and libraries including Scikit-learn, TensorFlow, and PyTorch for model development and experimentation.
- Skilled in data processing, feature engineering, exploratory data analysis, and transforming structured and unstructured data into machine learning-ready datasets.
- Hands-on experience developing supervised learning, unsupervised learning, NLP, computer vision, and forecasting models based on business requirements.
- Experienced in model evaluation, cross-validation, hyperparameter tuning, optimization techniques, and performance analysis.
- Knowledge of MLOps practices including model versioning, experiment tracking, containerized deployment, and machine learning lifecycle management.
- Familiar with cloud-based machine learning solutions using AWS services for model development, deployment, and experimentation.
- Experienced working with SQL databases, data pipelines, and analytical workflows for AI/ML applications.
- Strong understanding of model monitoring, retraining strategies, explainability techniques, and responsible AI practices including bias analysis and data privacy considerations.
- Experienced with Agile development methodologies, JIRA-based tracking, technical documentation, experiment reporting, and collaboration with cross-functional teams.

WORK EXPERIENCE:

AI/ML Engineer @ Tenet Healthcare | Dallas, TX

Sep 2025 – Present

- Developed machine learning solutions to support healthcare analytics initiatives, improving data-driven decision-making for clinical and operational processes.
- Performed data collection, validation, preprocessing, and exploratory analysis on healthcare datasets containing structured and unstructured information.
- Designed feature engineering pipelines to extract meaningful clinical, operational, and patient-related attributes for predictive modeling.
- Built supervised machine learning models for classification and prediction use cases using Scikit-learn.
- Applied statistical analysis techniques to identify patterns, trends, and relationships within healthcare data.
- Performed model training, cross-validation, and hyperparameter tuning to optimize machine learning model performance.
- Evaluated models using appropriate performance metrics and analyzed results to support model improvement decisions.
- Developed NLP workflows for processing healthcare documents and extracting meaningful information from text-based data.
- Implemented model explainability approaches to improve transparency and support responsible AI practices.
- Collaborated with healthcare analysts, data engineers, and business stakeholders in Agile development environments.
- Maintained technical documentation including model details, experiments, assumptions, and validation results.
- Supported deployment workflows by preparing machine learning artifacts and integrating models into application environments.
- Monitored model performance and participated in retraining discussions based on data changes and business requirements.
- Followed healthcare data privacy practices and maintained awareness of compliance requirements such as HIPAA.
- Used JIRA and Agile ceremonies to track development activities, enhancements, and model improvement tasks.

Technologies Used: Python, Scikit-learn, TensorFlow, Pandas, NumPy, SQL, AWS SageMaker, MLflow, Docker, JIRA.

Machine Learning Engineer @ American Express | New York, NY

Oct 2021 – Jul 2023

- Developed AI and machine learning solutions for financial services use cases including customer analytics, risk analysis, and predictive modeling.
- Prepared large datasets by performing data cleaning, transformation, validation, and exploratory data analysis.

- Created machine learning features from transactional and behavioral datasets to support predictive analytics applications.
- Built classification models to identify patterns and support business decision-making processes.
- Implemented machine learning experimentation workflows using Python and Scikit-learn.
- Applied hyperparameter optimization techniques to improve model stability and generalization.
- Conducted model evaluation using statistical metrics and analyzed model behavior across different datasets.
- Developed anomaly detection approaches to identify unusual patterns in financial data.
- Applied NLP techniques for processing customer-related text data and extracting relevant insights.
- Used MLflow for experiment tracking, model comparison, and maintaining reproducibility of machine learning workflows.
- Collaborated with data engineers to integrate machine learning pipelines with enterprise data platforms.
- Supported deployment activities by packaging models using Docker-based workflows.
- Participated in Agile sprint planning, technical discussions, and code reviews with engineering teams.
- Documented machine learning approaches, experiments, and model validation processes.
- Assisted in monitoring deployed models and identifying opportunities for retraining and optimization.

Technologies Used: Python, Scikit-learn, Pandas, NumPy, SQL, MLflow, Docker, AWS SageMaker, NLP, JIRA.

Junior Machine Learning Engineer @ Amazon | Goodlettsville, TN

Dec 2019 – Sep 2021

- Developed machine learning models for e-commerce analytics, customer behavior analysis, and recommendation-focused applications.
- Performed data preprocessing, feature extraction, and exploratory analysis on customer, product, and transaction datasets.
- Designed machine learning pipelines to prepare high-quality datasets for model training and evaluation.
- Developed recommendation and predictive modeling solutions using machine learning techniques.
- Built regression and classification models to analyze business patterns and customer interactions.
- Implemented feature engineering strategies to improve model input quality and support model experimentation.
- Trained and evaluated machine learning models using Scikit-learn and Python-based workflows.
- Applied cross-validation and hyperparameter tuning techniques to improve model reliability.
- Analyzed model outputs and generated insights to support business and product decisions.
- Worked with data teams to integrate machine learning components into analytical workflows.
- Used AWS services for storing data, running experiments, and supporting machine learning development activities.
- Applied MLflow practices for tracking experiments and maintaining model development history.
- Supported model deployment preparation using Docker and automated development workflows.
- Documented technical approaches, experiments, and model evaluation findings.
- Participated in Agile ceremonies and collaborated with engineers, analysts, and product teams.

Technologies Used: Python, Scikit-learn, Pandas, NumPy, SQL, AWS SageMaker, AWS S3, MLflow, Docker, JIRA.

TECHNICAL SKILLS:

- **Programming Languages:** Python, SQL, Java.
- **Machine Learning Frameworks:** Scikit-learn, TensorFlow, PyTorch.
- **Deep Learning Architectures:** Neural Networks, CNNs, Transformers, Autoencoders.
- **Data Processing:** Pandas, NumPy, Data Cleaning, Feature Engineering, Exploratory Data Analysis.
- **Machine Learning Techniques:** Regression, Classification, Clustering, Recommendation Systems, Time-Series Forecasting, Predictive Modeling.
- **Natural Language Processing:** Text Classification, Sentiment Analysis, Text Embeddings, Named Entity Recognition, Document Processing.
- **Computer Vision:** Image Classification, Image Preprocessing, Object Detection, Feature Extraction.
- **MLOps Tools:** MLflow, Docker, CI/CD Pipelines, Model Versioning.
- **Cloud ML Platforms:** AWS SageMaker, AWS S3, AWS Lambda.
- **Databases:** SQL Server, PostgreSQL, NoSQL Databases.
- **Visualization Tools:** Matplotlib, Power BI.
- **Methodologies:** Agile, Scrum, SDLC, Model Development Lifecycle.
- **Project Management Tools:** JIRA, Confluence.

EDUCATION:

Master of Science in Information Technology @ Kennesaw State University